WakeBand rigid rev A: 4-layer 0.8 mm FR4 board, about 15 x 33 mm, parts side down over a 60 mAh cell and an 8 mm LRA.
Cell: 60 mAh protected LiPo, about 16 x 13 x 4-5 mm, wire leads (X2 Robotics; alternate Ampul 401215). Pod about 17.5 x 35 x 9 mm.
Sheets connect through global labels and power nets. Same circuit as flex rev A; only the cell, the dock pad footprint and the layout changed.

power

Power: nPM1304 PMIC, battery, NTC, ship mode, current jumper

File: power.kicad_sch

mcu

MCU: ISP2454 (nRF54L15) module, SWD, UART, I2C, interrupts

File: mcu.kicad_sch

sensors

Sensors: LIS2DW12 accelerometer

File: sensors.kicad_sch

haptics

Haptics: DRV2605L driver and LRA pads

File: haptics.kicad_sch

dock

Dock: six pogo pads, TVS, series resistors

File: dock.kicad_sch

debug

Debug: test pads, debug LED (DNP option)

File: debug.kicad_sch

Plan: docs/flex-pcb-schematic-plan.md

FOSS Health Alarm

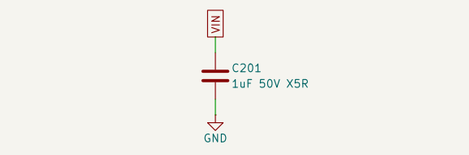
Sheet: /

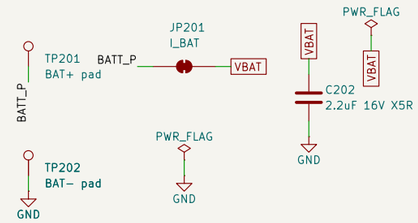
File: rigid-rev-a.kicad_sch

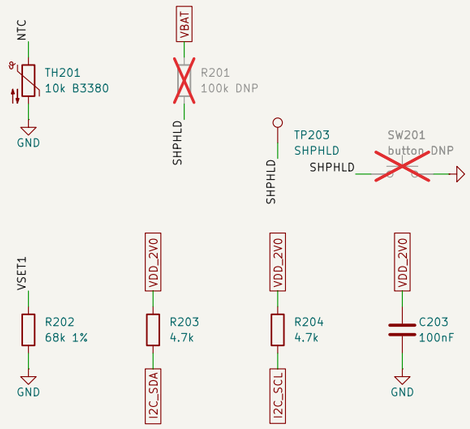
Title: WakeBand rigid rev A — Top

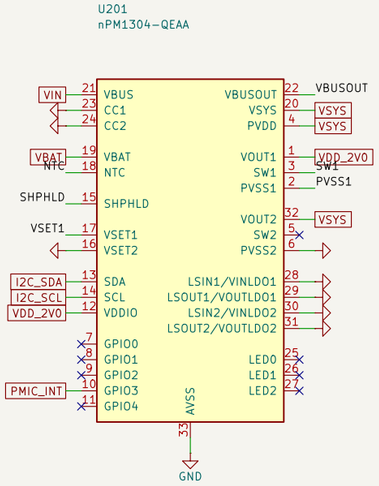
Size: A4	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 60 m**
KiCad E.D.A. 10.0.5		Id: 1/7

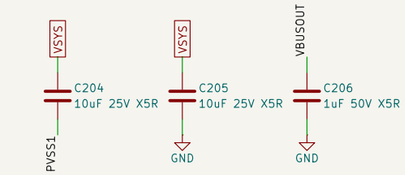
nPM1304 power sheet (step 3). Based on Nordic nPM1304 datasheet, reference circuit 'Configuration 2' (QFN):
one buck, no LDO/load switch, ship-mode exit and NTC used. Draft until the Stage 0b nPM1304-EK tests.

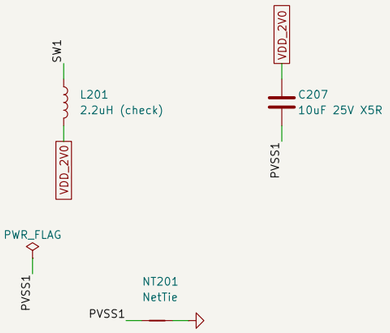












Strapping and connections (from the datasheet):
- VBUS from dock pad VIN. VBUS abs max 22 V: OVP 5.5 V typ; VBUSPOR 3.9 V. The VIN TVS (D601) is on the dock sheet, at the pad.
- CC1/CC2 to GND: no USB-C; VBUS limit stays at the 100 mA default (enough for the 30 mA charge current).
- VBUSOUT not used by the SoC but needs its 1 uF decoupling cap.
- VSET1 = 68k -> BUCK1 starts at 2.0 V (datasheet table: 68k = 2.0 V, max 5% tolerance). VOUT1 = VDD_2V0.
  2.0 V (not 1.8 V) leaves margin over the ISP2454 1.7 V minimum when the buck is 5 % low (1.9 V).
- BUCK2 unused: VSET2 to GND (off), VOUT2 to VSYS, SW2 open, PVSS2 to GND, no inductor (as in reference config 2).
- LOADSW1/2 unused: LSIN1, LSOUT1, LSIN2, LSOUT2 to GND (as in reference config 3). LED0-2 open.
- VDDIO from VOUT1. Unused GPIO0/1/2/4 open: default is input with weak pull-down.
- PMIC_INT on GPIO3 (any GPIO can be the interrupt; set in devicetree).
- PVSS1 joins GND through net tie NT201 (one via to the GND layer), as in the reference layout.
- NTC: 10k 1% B25/50 = 3380 K, one of the three supported types; JEITA defaults match it. Max 100 pF across it.
- Ship mode exits on SHPHLD low (button to GND) OR VBUS > VBUSPOR, so docking alone wakes the band
  (still confirm on the EK in Stage 0b).
- DECIDED 2026-10-30: SHPHLD relies on the nPM1304's internal 50k pull-up. R201 (extra pull-up to VBAT) and
  SW201 (button) stay DNP; TP203 is the way to pull SHPHLD low on the bench.
- TWI address 0x6B. I2C pull-ups R203/R204 serve the whole 2.0 V bus (4.7k for 400 kHz).
- JP201: cut to put the PPK2 in series with the cell; bridge again with solder.
- CELL (rigid rev A): 60 mAh protected LiPo, ~16 x 13 x 4-5 mm, wire leads to TP201 (+) / TP202 (-).
  Source X2 Robotics (x2robotics.ca, 3.7 V 60 mAh); alternate Ampul 401215 (4 x 12 x 15 mm).
  Charge at <= 30 mA (0.5C). nPM1304 power-on defaults: charging disabled, ICHG 4 mA, VTERM 4.0 V.
  Firmware must set ICHG = 30 mA and VTERM = 4.2 V over I2C, then enable charging.

(check) before freezing:
- U201 footprint: KiCad QFN-32 5x5 EP 3.45 mm; datasheet EP is 3.5 nom (3.4-3.6). Compare with Nordic's QFN reference layout.
- U201 has no LCSC stock (only the nPM1304-EK, C50090302); source the chip elsewhere or as a consigned part.
- L201: DFE201610E-2R2M on the KiCad DFE201610P land pattern; confirm the pad sizes.
- GPIO3 for PMIC_INT: match the nPM1304-EK / Zephyr devicetree used in Stage 0b.
- VSYS abs max 5.5 V: dock supply must stay at 5.0 V; DRV2605L blocked while VBUS is present.

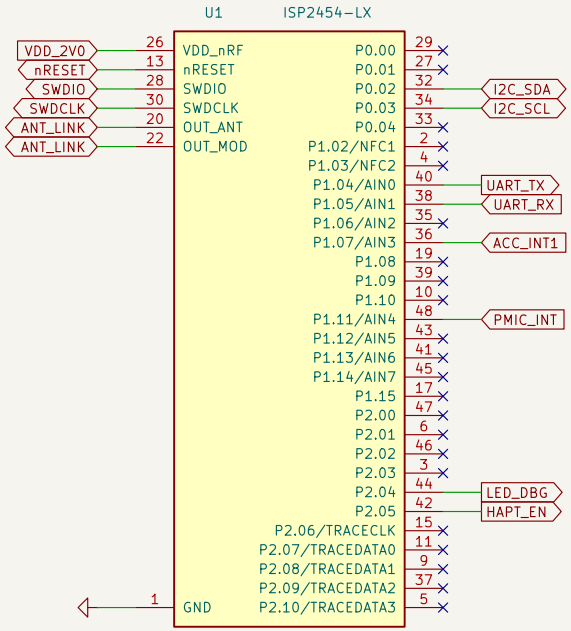
nPM1304 per Nordic reference circuit configuration 2 (QFN32)
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

Sheet: /power/
File: power.kicad_sch

Title: WakeBand rigid rev A — Power

Size: A3	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 60 r**
KiCad E.D.A. 10.0.5		Id: 2/7

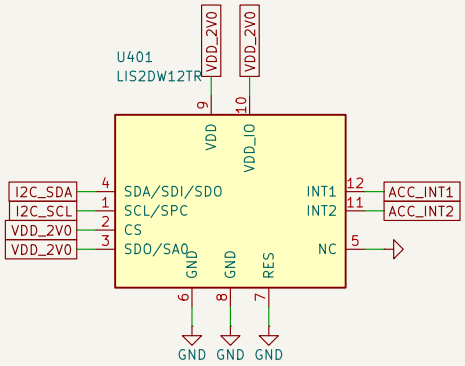
ISP2454 notes (datasheet R7):
- Antenna, 32 MHz + 32.768 kHz crystals, DC-DC inductor and all decoupling are inside. No external parts.
- OUT_ANT (20) must be joined to OUT_MOD (22) by a short trace (net ANT_LINK).
- Pads 49-62 are mechanical: isolated copper in the footprint, not in this symbol.
- Antenna corner 1 mm from the board edge, no copper or parts on any layer in the keep-out, and keep the cell and LRA 2 mm (plan view) clear of it.
- Firmware: set LFXO load capacitance to 19 pF (devicetree load-capacitance-femtofarad = <19000>).
- Signals use OUTER-row pads. Rigid rev A: nRESET (13) and the trapped GND pads (12, 14, 16, 18, 21, 23) use epoxy-filled, copper-capped vias in pad (JLCPCB POFV). Pad 10 stays NC.
- I2C on TWIM30 (P0, low-power domain). UART on UARTE20 (P1).
- P1.02/P1.03 are NFC pins: left unused. P2 trace pins left unused.
- GPIO plan and reasons: docs/flex-rev-a-pinout.md

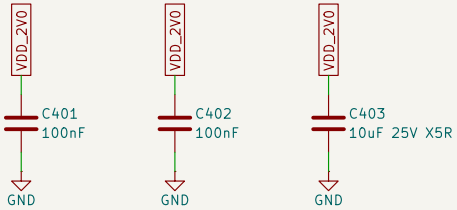


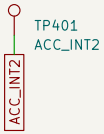
Review 2026-10-30:
- Pad 10 (P1.10) is not connected: it sits in the lower outer row, which cannot be routed without via-in-pad (antenna keep-out below, 0.25 mm gaps to the inner pads). The DRV2605L IN/TRIG is tied to GND; effects run over I2C.
- GND pads 12, 14, 16 and 18 sit in the trapped lower row. They are tied together inside the module. Rigid rev A ties each one to the L2 GND plane with a filled, capped via in pad (POFV). Pads 8 and 24 reach the large pads 7 and 25 by a short trace. All of them stay on the GND net in the schematic.
- VDD_nRF is VDD_2V0 (BUCK1 at 2.0 V, margin over the 1.7 V minimum when the buck is 5 % low).

Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /mcu/
File: mcu.kicad_sch
Title: WakeBand rigid rev A — MCU
Size: A4 | Date: 2026-11-02 | **Rev: RA0 (rigid 4-layer, 60 r**
KiCad E.D.A. 10.0.5 | Id: 3/7

Sensors sheet (step 5): ST LIS2DW12 on the shared 2.0 V I2C bus. Pinout from ST datasheet DocID029682 Rev 4, table 2.







- VDD (pin 9) and VDD_IO (pin 10) on VDD_2V0; C401 at pin 9 (VDD), C402 at pin 10 (VDD_IO).
- C403 10 uF bulk on VDD near pin 9, per ST's application hint (100 nF + 10 uF).
- CS (pin 2) tied to VDD_2V0: I2C mode (SPI disabled).
- SDO/SA0 (pin 3) tied to VDD_2V0: I2C address 0011001b = 0x19 (0x18 if SA0 were low).
  CS and SA0 also have internal pull-ups (about 49k at 1.8 V, 30k at 2.5 V); the hard strap avoids depending on them.
- SDA/SCL on I2C_SDA/I2C_SCL. No pull-ups here: R203/R204 on the power sheet serve the whole bus.
- INT1 -> ACC_INT1 (SoC wake GPIO P1.07). INT2 -> ACC_INT2, test pad TP401 only.
- Pin 5 NC: datasheet allows VDD, VDD_IO or GND; tied to GND. Pin 7 RES: connect to GND (datasheet).
- Pins 6 and 8 GND.

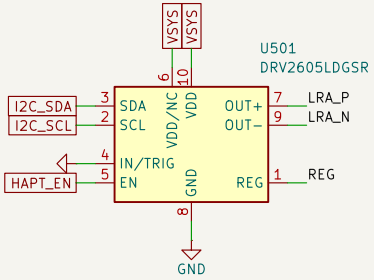
LIS2DW12 per ST datasheet DocID029682 Rev 4
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

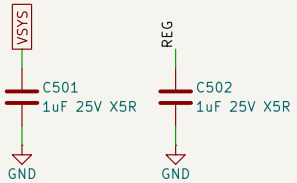
Sheet: /sensors/
File: sensors.kicad_sch

Title: WakeBand rigid rev A — Sensors

Size: A4	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 60 r**
KiCad E.D.A. 10.0.5		Id: 4/7

Haptics sheet (step 5): TI DRV2605L (VSSOP-10, DGS) on VSYS, I2C at 2.0 V. From TI datasheet SLOS854D.







- VDD (pin 10) from VSYS with C501 1 uF; pin 6 VDD/NC tied to VDD as the datasheet allows.
- REG (pin 1): internal 1.8 V regulator output, C502 1 uF required.
- EN (pin 5) <- HAPT_EN (P2.05).
- IN/TRIG (pin 4) tied to GND (datasheet: connect to GND if not used). Effects are triggered over I2C (GO bit). HAPT_TRIG was dropped: ISP2454 pad 10 (P1.10) is in the lower outer row and cannot be routed without via-in-pad.
- SDA/SCL on the shared bus; VIH = 1.3 V, so 2.0 V logic is fine. I2C address 0x5A.
- OUT+ (7) / OUT- (9) to the LRA lead pads TP501 (LRA+) and TP502 (LRA-), 1.0 x 1.5 mm, hand-soldered.
- MECHANICAL (rigid rev A): the 8 mm LRA is glued to the case floor under the board, not a schematic part. Motor: Vybronics VG0832012. Leads hand-soldered to TP501/TP502 on F.Cu.
- DO NOT switch the driver's VDD off: SDA, SCL, EN and IN/TRIG are limited to VDD + 0.3 V, so an unpowered DRV2605L would be back-powered through its I2C pins and clamp the shared bus. Rev A keeps VDD on with EN low (about 4 uA). Firmware blocks the motor while VBUS is present (VSYS then follows VBUS up to 5.5 V, above the 5.2 V recommended maximum; abs max is 5.5 V).

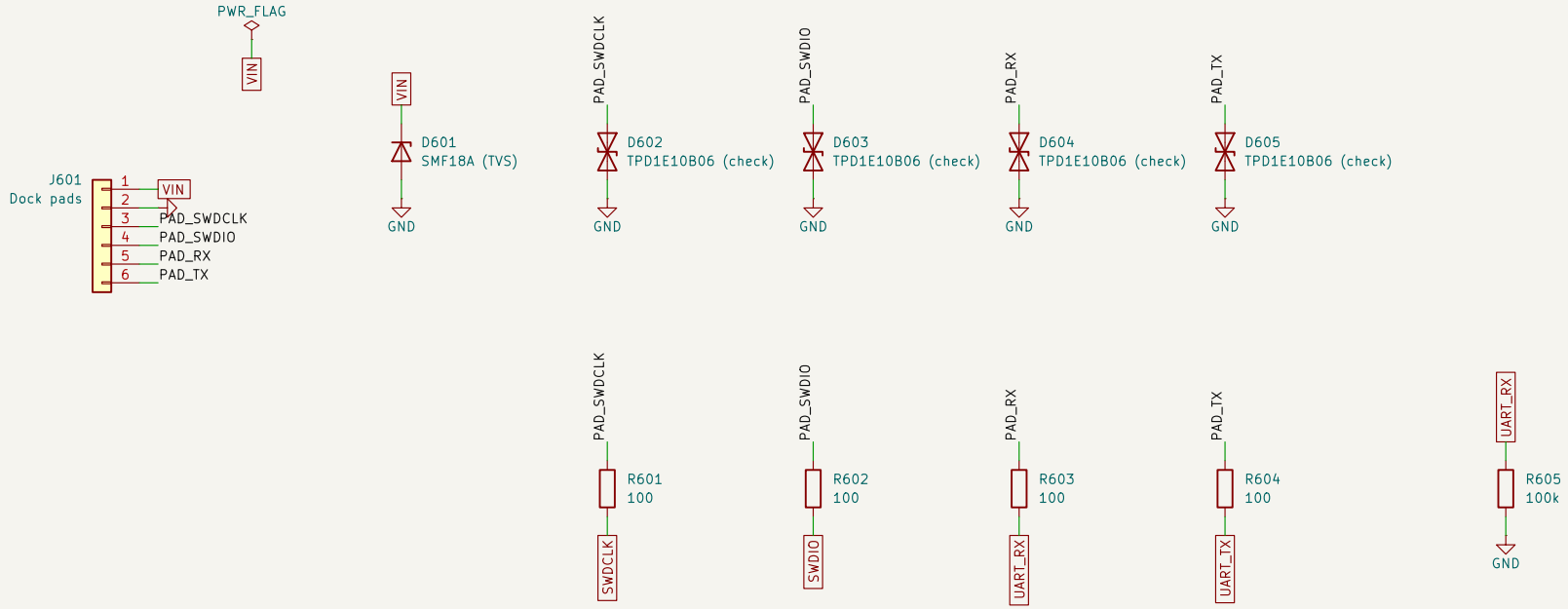
DRV2605L per TI datasheet SLOS854D
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

Sheet: /haptics/
File: haptics.kicad_sch

Title: WakeBand rigid rev A — Haptics

Size: A4	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 60 r**
KiCad E.D.A. 10.0.5		Id: 5/7

Dock sheet (step 6): six round 1.6 mm pogo pads on B.Cu (the pod outer face), 2.5 mm pitch (J601 footprint, rigid rev A).



Pad order: 1 VIN, 2 GND, 3 SWDCLK, 4 SWDIO, 5 UART_RX, 6 UART_TX.
- TVS at the pad on VIN (D601) and on each data pad (D602–D605). No TVS on pad 2: it is the GND reference.
- D601 is Littelfuse SMF18A (SOD-123FL, unidirectional, 18 V standoff, 20.0–22.1 V breakdown, 1 uA max at 18 V), so a wrong 9–12 V supply does not burn it. The nPM1304 VBUS tolerates 22 V and cuts VSYS off above 5.5 V. D601 replaces the VIN TVS that was on the power sheet (the pads and the PMIC are on different islands).
- 100 ohm in series between each data pad and the SoC net (R601–R604), TVS on the pad side.
- R605 100k pull-down on UART_RX at the SoC side, so the RX input does not float off the dock.
- Firmware enables the UART only while the PMIC reports VBUS present.
- SWDIO has an always-on internal pull-up in the nRF54L15, so pad 4 sits at 2.0 V on the skin all night.
- Pads: ENIG for rev A (JLCPCB rigid); hard gold (PCBWay) if pogo wear shows.

Pad order from docs/hardware-roadmap.md
Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm

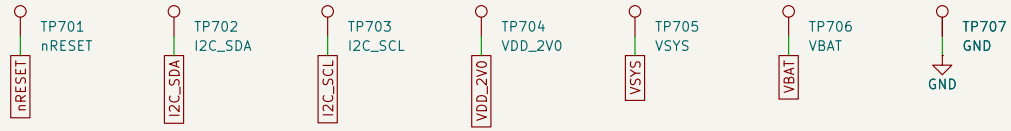
Sheet: /dock/
File: dock.kicad_sch

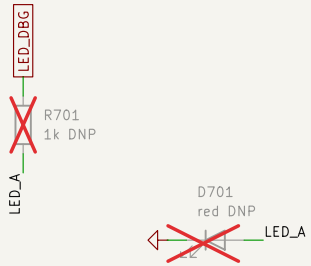
Title: WakeBand rigid rev A — Dock

Size: A4	Date: 2026-11-02	**Rev: RA0 (rigid 4-layer, 60 r**
KiCad E.D.A. 10.0.5		Id: 6/7

(check):

Debug sheet (step 6): 0.8 mm round test pads (no paste) on F.Cu, and a DNP debug LED.





- Test pads: nRESET, I2C_SDA, I2C_SCL, VDD_2V0, VSYS, VBAT, GND (TP701–TP707). SHPHLD (TP203, power sheet) and ACC_INT2 (TP401, sensors sheet) already have pads.
- nRESET: test pad only. The nRF54L15 has an internal pull-up on nRESET, and SWD can reset the chip, so nothing on the board drives this input. ERC: the passive test pad satisfies KiCad's 'input not driven' check, so the earlier nRESET error is gone without any exclusion. This is intended, not a missing driver.
- LED_DBG (P2.04) -> R701 1k -> D701 (0402 red) -> GND, both DNP. Fit them only for bench bring-up: a lit LED would cost far more than the sleep budget.

Plan: docs/flex-pcb-schematic-plan.md
FOSS Health Alarm
Sheet: /debug/
File: debug.kicad_sch
Title: WakeBand rigid rev A — Debug
Size: A4 | Date: 2026-11-02 | **Rev: RA0 (rigid 4-layer, 60 r**
KiCad E.D.A. 10.0.5 | Id: 7/7